

PROGRAMME INDEX NUMBER

**KWAME NKRUMAH UNIVERSITY OF SCIENCE AND
TECHNOLOGY, KUMASI**

INSTITUTE OF DISTANCE LEARNING

DIPLOMA IN COMPUTER NETWORKING MIDSEM EXAMINATION, 2018

EE 156 : ELECTRICAL ENGINEERING DRAWING

(First Year)

November , 2018

One Hour

INSTRUCTIONS FOR CANDIDATES:

1. Answer **All** questions
2. Indicate your ***index number*** on top of the question paper

1. The appropriate cable size for lighting and fans is
 a) 1.5mm² b) 2.5mm² c) 4.0mm² d) 6.0mm²
2. In electrical services design, how many distribution boards will be ideal for 12 circuits
 a) Two 6-way DBs b) Two 8-way DBs c) Three 6-way DBs d) Three 8-way DBs
3. The typical power rating of a ceiling fan is
 a) 40W b) 80W c) 100W d) 60W
4. The benefit of using CAD in drawing includes
 a) Saving time in the preparation of all types of electrical drawing
 b) Eliminating tedious tasks of lettering and drawing uniform lines and symbols
 c) Making rapid corrections of mistakes or oversights that have been discovered.
 d) All of the above

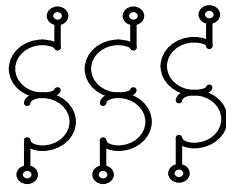


Figure 1

5. The symbol shown in figure 1 above is
 a) an overload coil b) an overload contact c) a fuse link d) a disconnect switch

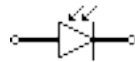


figure 2

Use Figure 2 above to answer questions 6 and 7.

6. Figure 2 is the symbol for
 a) Led b) varactor c) tunnel diode d) photo diode
7. The best description of the operation of the element in figure 2 is
 a) it allows current to flow when exposed to light
 b) it emits light when current flows through it
 c) it is a variable capacitance diode
 d) all of the above
8. To control lamps along a long hallway it will be preferable to use which of the following kinds of switches
 a) two-way b) two-gang c) three-gang d) single switch

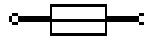


Figure 3

9. Figure 3 is best described as a/an
 a) switching device **b) protection device** c) electrical machine d) current transformer
10. The 2-way home-run symbol represents _____ wires.
 a) live and earth b) earth and neutral **c) live and neutral** d) live, earth and neutral

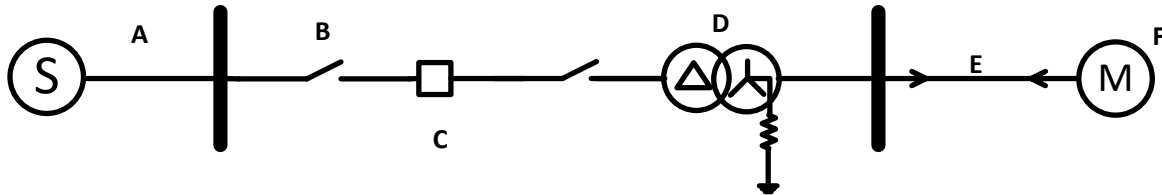


Figure 4

Use the figure above to answer questions 11 to 17

11. In figure 4 ,the portion labelled A is
a) an infinite bus b) a dc generator c) a synchronous motor d) none of the above
12. Which of the labelled symbols represents an isolator?
a) B b) C c) E d) F
13. The switching devices found in Figure 4 is/are I) B II) C III) E
 a) I only **b) I and II only** c) II and III only d) I, II and III
14. Which of the elements in figure 4 is a protective device
 a) C b) B **c) E** d) D
15. The element labelled C in figure 4 can be described as
 I) a switching device II) a protective device III) a power source
 a) I only **b) I and II only** c) II and III only d) I, II and III
16. Element D in figure 4 is a
 a. Delta-Star solidly grounded single-phase transformer
 b. Delta-Star resistance grounded single phase transformer
c. Delta-Star solidly grounded three phase transformer
 d. Delta-Star reactance grounded single phase transformer
17. The motor is connected to the system with the help of
 a) a lightning arrestor b) an overhead conductor **c) underground cables** d) general wires



Figure 5

18. Figure 5 is a symbol for

- a)Earth ground b)Digital ground c)Common ground d)Chassis ground

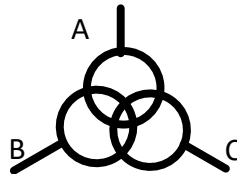


Figure 6

Use figure 6 to answer questions 19 and 20.

19. The symbol shown in figure 6 above is a

- a.three-phase three winding autotransformer
b.single-phase three winding autotransformer
c.three-phase double winding transformer
d.single-phase double winding transformer

20. The terminals labelled A, B and C in figure 6 are _____, _____ and _____ windings respectively.

- a)LV, HV, MV b)HV, LV₁, LV₂ c)MV, LV, HV d)HV, MV, LV

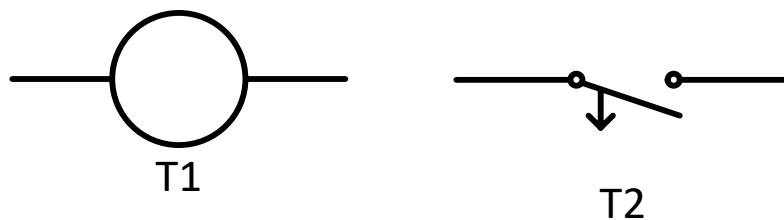


Figure 7

Figure 7 above shows a coil and its corresponding contact. T1 controls T2.

Use the information to answer questions 21 to 22

21. Symbol T1 shown in figure 7 above is a/an

- a)On-delay timer b)Off-delay time c)Overload relay d)Control relay

22. In figure 7, if the coil T1 is energized what happens to T2?
 a.) Nothing happens **b) T2 immediately closes** c) T2 closes after a delay
 d) T2 becomes Normally Closed
23. In figure 7, if the coil T1 is de-energized
 a) Nothing happens b) T2 opens immediately **c) T2 opens after a delay**
 d) T2 becomes Normally Closed
24. In a motor control circuit, the stop button is usually
a) a normally closed push button b) a normally opened push button
 c) a selector switch d) a toggle switch

Figure 8 below shows symbols (A to F) that are used to represent certain switches in motor control schematic diagrams. *Use it to answer questions 25 - 26*

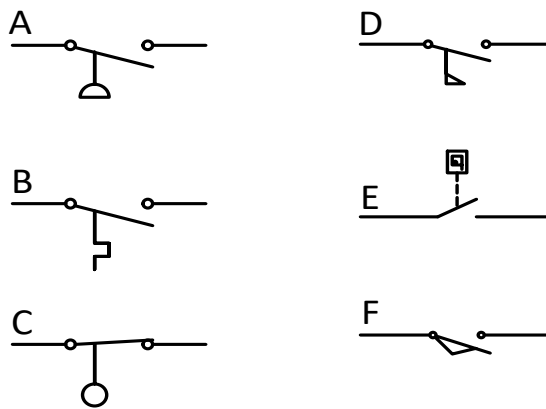


Figure 8

25. Which symbol in figure 8 represents a switch designed to pass electrical signal only when a predetermined limit is reached?
 a. D **b) F** c) B d) C
26. The temperature actuated switch in figure 8 is
 a) A **b) B** c) E d) C
27. Which switch in figure 8 monitors the direction of movement of liquid or gas substances
b. D b) F c) A d) E

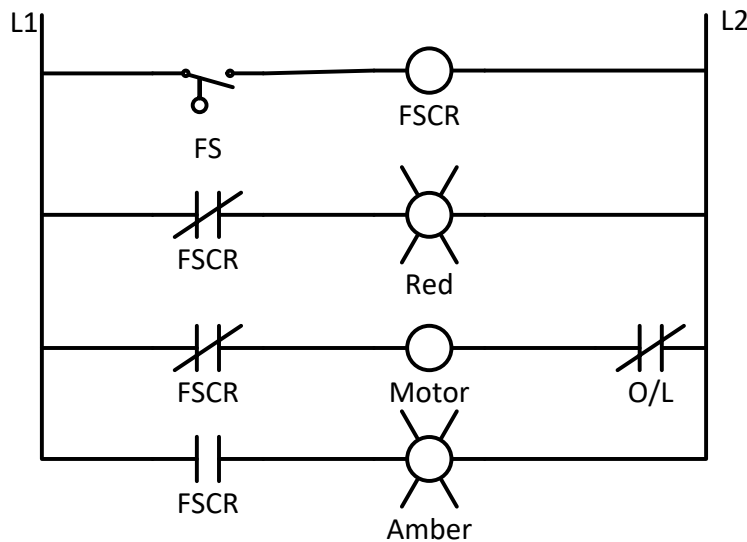


Figure 9

Figure 9 above is a control circuit diagram for a pump that fills a tank with water.

Use it to answer questions 28-34.

28. Which of the following devices comes on when the tank is empty?

- a) Red lamp b) Amber lamp c) Motor d) Both a and c

29. When the red lamp is on, it indicates that

- I) The tank is full II) The tank is not full III) The motor is on IV) The switch FS is closed
a) I and III only b) II and III only c) III and IV only d) I, III and IV only

30. The amber lamp comes on when

- a) the overload trips b) FS closes c) the motor is running d) FSCR de-energizes

31. FS closes when

- c.) the tank is full b) the tank is empty c) the o/l relay trips d) FSCR energizes

32. The symbol labelled motor is, in reality, a/an

- a) Control relay b) Contactor coil c) Overload coil d) Timer relay

33. When excessive current passes through the motor

- a) the red lamp blinks b) the amber lamp blinks c) O/L trips d) FSCR de-energizes

34. If the control circuit at figure 10 were to be numbered, the terminals of O/L will be numbered

- a) 7,8 b) 7,3 c) 8,9 d) 6,3

35. The symbol labelled O/L is an overload

- a)coil position b)relay contact c)main contact d) push button

36. In Single-Line Diagrams, it is assumed that all the three phases of a power system are

- a)balanced b) unbalanced c)equal d)disorganized

37. A two-way switch operates by allowing a user to

- a)control two sets of lamps from the same location.
b)control one set of lamps from two different locations.
c)switch on lamps from one location and then switch them off at a different location.
d)All of the above

38. In order to connect sockets into circuits, sockets within what maximum area should be connected in a ring circuit?

- a)50m² b)150m² c)100m² d)None of the above

39. Why are lamps connected in parallel?

- a)To ensure that all the lamps receive the same supply voltage.
b)To reduce the total power taken from the source.
c)To improve the power factor of the lighting circuit.
d) To reduce the total current drawn from the source.

40. The number of 40W incandescent lamps that can be connected in one circuit is

- a)13 b)12 c)8 d)7

